

IBRA, Oman

WMO: 412650

Lat: 22.8094N

Lon: 58.4606E

Elev: 1539

StdP: 13.90

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.7	56.1	19.5	15.5	80.1	23.6	18.9	81.0	17.6	72.8	14.2	73.9	3.5	320	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	23.8	111.7	70.4	110.0	70.7	108.2	70.9	79.9	93.5	79.0	93.4	78.2	92.8	9.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	150.5	84.7	75.8	142.9	84.9	74.7	137.5	85.1	45.0	91.9	44.0	93.4	43.1	93.0	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.1	13.1	DB	48.3	115.5	2.2	1.4	46.8	116.5	45.5	117.3	44.3	118.1	42.7	119.1	(4)
(5)			WB	38.7	80.6	2.7	1.3	36.8	81.6	35.2	82.3	33.7	83.0	31.7	83.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	67.8	71.2	78.1	86.0	92.8	95.7	95.1	93.2	90.1	84.7	76.3	70.3	(6)	
	DBStd	10.57	3.26	4.41	4.24	4.00	3.36	3.78	4.39	4.31	3.38	3.16	3.32	3.45	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	21	12	5	0	0	0	0	0	0	0	0	0	4	(9)	
	CDD50	12229	551	594	872	1081	1326	1371	1399	1340	1203	1074	790	628	(10)	
	CDD65	6776	98	179	407	631	861	921	934	875	753	609	340	168	(11)	
	CDH74	96967	740	1739	4684	8831	13783	15231	14866	13308	10943	8071	3423	1348	(12)	
	CDH80	61674	97	553	2341	5367	9418	10942	10502	9056	7000	4707	1407	284	(13)	
(14) Wind	WSAvg	5.4	4.5	5.5	5.8	6.2	6.4	6.5	5.8	5.6	5.3	4.9	4.5	4.3	(14)	
(15) Precipitation	PrecAvg	5.2	0.2	0.7	0.8	1.0	0.2	0.4	0.4	0.4	0.2	0.2	0.4	0.3	(15)	
	PrecMax	9.1	2.1	4.6	4.7	3.9	1.0	3.9	1.9	2.0	1.8	2.4	3.8	2.6	(16)	
	PrecMin	0.5	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	2.7	0.5	1.2	1.2	1.1	0.3	0.9	0.6	0.4	0.4	0.6	0.9	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.6	91.8	98.6	104.5	112.7	114.3	114.1	111.5	108.6	102.8	94.2	88.3	(19)	
		MCWB	59.5	62.0	63.4	66.4	69.6	69.1	72.5	71.7	69.6	65.7	65.2	61.6	(20)	
	2%	DB	82.6	89.0	96.0	102.2	108.6	111.9	111.3	109.4	106.0	100.5	91.6	85.3	(21)	
		MCWB	61.0	62.0	63.3	66.3	69.2	69.1	72.0	72.2	69.7	65.9	63.9	60.7	(22)	
	5%	DB	80.3	86.1	93.5	100.1	106.2	110.2	109.3	107.6	103.9	98.5	89.6	83.2	(23)	
		MCWB	60.8	61.8	63.1	65.7	68.9	69.3	72.4	72.2	70.1	65.5	63.2	61.1	(24)	
	10%	DB	78.2	83.4	91.1	98.1	104.1	108.0	107.3	105.5	101.7	96.3	87.6	81.0	(25)	
		MCWB	60.3	61.5	62.8	65.4	68.8	69.5	72.8	73.1	70.2	64.8	63.2	61.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.7	70.2	72.3	75.7	79.2	81.5	80.9	80.3	78.9	75.0	72.8	69.0	(27)	
		MCDB	76.1	77.3	80.1	83.5	89.0	90.6	97.7	95.1	93.6	85.9	84.3	75.5	(28)	
	2%	WB	65.9	67.5	70.7	73.4	77.9	80.0	79.7	79.2	77.1	73.6	70.0	67.5	(29)	
		MCDB	73.8	76.5	79.1	84.2	88.5	90.4	96.3	94.9	90.5	84.3	78.2	74.6	(30)	
	5%	WB	64.5	66.2	68.8	72.0	76.9	78.9	78.8	78.5	76.1	72.4	68.6	66.3	(31)	
		MCDB	73.2	75.3	80.2	84.1	88.2	91.0	95.0	94.4	90.3	84.0	78.0	74.2	(32)	
	10%	WB	63.1	64.7	67.1	70.6	75.6	77.7	77.9	77.4	75.0	70.9	67.3	65.0	(33)	
		MCDB	72.4	76.0	80.6	85.3	89.5	91.2	93.9	93.2	90.2	84.0	78.1	74.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	22.9	24.0	23.2	23.3	23.8	23.3	23.5	23.7	23.3	21.7	21.2	(35)	
		MCDBR	24.2	26.3	26.9	24.9	25.0	26.0	24.8	24.8	24.9	25.1	23.7	24.1	(36)	
	5% WB	MCWBR	8.2	8.6	8.5	6.8	7.2	7.5	5.9	5.3	6.5	7.1	8.2	8.1	(37)	
		MCWBR	20.3	21.6	20.9	21.7	23.0	23.6	23.0	23.1	23.9	21.6	20.0	18.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.381	0.421	0.480	0.542	0.582	0.659	0.737	0.676	0.561	0.439	0.404	0.379	(40)	
		taud	2.293	2.143	1.955	1.835	1.753	1.625	1.528	1.621	1.814	2.097	2.232	2.295	(41)	
	Ebn at Noon		278	273	262	248	236	217	201	213	237	264	268	274	(42)	
		Edh at Noon	38	47	59	68	73	83	91	83	67	49	40	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		1499	1753	2029	2243	2426	2333	2208	2142	2055	1869	1563	1419	(44)	
	RadStd		59	65	102	94	41	93	105	60	38	46	43	58	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		+0.41	N/A	N/A	N/A	+0.70	+0.81	N/A	N/A	+173	+176	

Nomenclature: See separate page